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BACHELOR THESIS

An Exponential Upper Bound on Probabilities of Atypical Sets of Discrete
Memoryless Sources

Alexander Jerschow

Bachelor of Science (BsC)

Vienna, 02-2024

Student ID Number:

1204 9824

Field of Study:

Bachelor of Science Mathematics

Advisor:

Pf. Mathias Beiglböck

Abstract

This thesis serves as an introduction to entropy and types as a motivation for Sanov's inequality: an upper bound on probabilities of atypical sets that decays at an exponential rate. We draw a contrast to more well-known bounds like ones derived from Chebyshev's inequality, which decay to zero sublinearly. After introducing types and type classes, proving a few useful bounds and equalities involving their probabilities and cardinalities, we provide an intuitive interpretation of entropy as a measure of uncertainty and connect this interpretation to the asymptotic equipartition property. We then provide our aforementioned upper bound via Sanov's inequality and its interpretation as well as a lower bound, yielding a limit of mainly theoretical interest (Sanov's large deviation theorem). We conclude by providing a few examples of applications of this inequality to rare event analysis.

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1 Introduction

In many situations in statistical and probabilistic modelling, one wishes to minimize the probability of some error or event. For instance, often times it is the case that one is looking at some estimator, such as the empirical mean, of n independent and identically distributed (i.i.d.) random variables. Through a theoretical result such as the law of large numbers or otherwise, one usually has the convergence of this estimator towards some desired value. Aside from this convergence, one also wants to know how many trials one needs to guarantee a certain confidence of being within an ϵ of this desired value. This is one of the central goals of large deviations theory and rare event analysis.

In the case of the empirical mean, one possibility for placing an upper bound on the probability of being away from the true expected value could be found through Chebyshev's inequality:

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\text{Var}[\bar{X}_n]}{\epsilon^2}$$

where $\mu := E[X_1]$. We can use additivity of variances of independent random variables, the property that the variance is homogenous with degree 2 and that our random variables are identically distributed to attain:

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\text{Var}[X_1]}{n\epsilon^2}$$

Apart from the necessity of the existence of the second moment of our distribution, we also see that this error converges to zero sublinearly in n . In situations where carrying out trials to improve an estimator is expensive, with respect to time, money or some other resource, a tighter bound on such probabilities would obviously be more ideal, as it leads to a more resource-efficient solution to a problem. Our goal will be to find an upper bound on probabilities like these that decays exponentially.

2 Entropy

Our framework will be similar to as in the introduction. More specifically: a discrete memoryless source (DMS) is a sequence $(X_1, X_2, \dots)$ of independent and identically distributed (i.i.d) random variables taking values in a finite set E called the source alphabet. The underlying distribution of each X_i is called the generic distribution [5]. Moreover, one of the central concepts we will need is the entropy of a random variable:

$$H(X) := - \sum_{x \in E} p(x) \log p(x) \tag{2.1}$$

$$H(X) := - \int_E p(x) \log p(x) dx = - \int_{\Omega} \log p(X) dP = E[-\log(X)] \tag{2.2}$$

where Ω is our sample space, P the corresponding probability measure, and p the probability mass function or density of the random variable X . We will adopt the convention: $0 \cdot \log 0 = 0$. Additionally, (2.1) only applies for discrete X and the first integral of (2.2) only for continuous X . We may also write $H(Q)$ with Q being X 's distribution. It should be remarked that, depending on the context, the base of the logarithm is usually either e or 2. In this thesis we will use 2 unless explicitly stated otherwise but most results hold for arbitrary bases.

Entropy is a type of measure for uncertainty, information or "surprise". The idea is that less likely events should yield more "surprise" when they occur, which is why a function like $-\log$ with a pole at zero might be a natural choice. Indeed, we would then have a function that awards a large amount of "surprise" to an outcome x with small probability $p(x)$. Then, taking the weighted sum/integral of these logarithms (with weights allocated by p), one essentially gets the expectation, or average surprise of the outcome of a random variable. There is another motivation connected to the multinomial coefficient, which is the number of permutations of n elements, not distinguishing between n_1 elements, n_2 other elements, $n_3 \dots$ and is given by:

$$\binom{n}{n_1, \dots, n_k} = \frac{n!}{n_1! \cdots n_k!}$$

One can check that this is the number we seek by observing that every permutation that doesn't distinguish between n_1 elements yields $n_1!$ more permutations that do. Thus, we have $\frac{n!}{n_1!}$ possible permutations in the case $i = 1$ (continue inductively). It should be noted, this coefficient only makes sense if $n = \sum_{i=1}^k n_i$. We will see how, exactly, this number is connected with entropy after first discussing a few results relating to types:

3 Types and Type Classes

The type P_x of a sequence $x \in E^n$ produced according to our DMS is its histogram :

$$P_x(a) := \frac{N(a|x)}{n}$$

where $N(a|x) := |\{i \in [1, n]_{\mathbb{N}} | x_i = a\}|$ denotes the number of occurrences of the element $a \in E$ in our given sequence x . Additionally, the type class $T(P)$ of a given distribution $P \in \mathcal{P}_n$ is the set of sequences with type P :

$$T(P) := \{x \in E^n | P_x = P\}$$

where $\mathcal{P}_n$ denotes the distributions with denominator n , in other words, the set of distributions P , such that $\text{Im}(nP) \in \mathbb{N}$. Let us look at a short example illustrating these terms [2]:

$E := \{1, 2, 3\}, x := (1, 1, 1, 3, 2) \in E^5$, then:

$P_x(1) = 3/5, P_x(2) = P_x(3) = 1/5$, and:

$T(P_x) = \{x \in E^5 | N(3|x) = N(2|x) = 1, N(1|x) = 3\}$, the set of all $x \in E^5$ with 1 occurring three times, and 2 and 3 once

P_x can be thought of as the best guess for the underlying generic distribution that led to x and $T(P)$ the set of all sequences that would lead to a best guess of P . It can also be remarked, that the type classes form a partition of the space E^n . This is because every sequence $x \in E^n$ has a type given by the definition, thus is in a type class. Moreover, x cannot be in a different type class because then x would have two types and not be a permutation of itself (from the definition it is clear, that all permutations of a sequence lead to the same type as $N(a|x)$ is permutation-invariant), which is of course false. We will see that the set of sequences with type "close" to the generic distribution is where almost all of the mass of our (product) probability measure lies as $n \rightarrow \infty$. But first we need to specify what we mean when we say a distribution is "close" to another:

4 Kullback-Leibler Divergence

The Kullback-Leibler divergence (also called the relative entropy or informational divergence) will prove to suit our needs as a measure of similarity between two distributions:

$$D(P||Q) := \begin{cases} \sum_{x \in \Omega} P(x) \log \frac{P(x)}{Q(x)} & P \ll Q \\ \infty & \text{otherwise} \end{cases} \quad (4.1)$$

$$D(P||Q) := \begin{cases} \int_{\Omega} p(x) \log \frac{p(x)}{q(x)} dx = \int_{\Omega} \log \left(\frac{dP}{dQ} \right) dP & P \ll Q \\ \infty & \text{otherwise} \end{cases} \quad (4.2)$$

where (4.1) is meant for discrete distributions, (4.2) for continuous ones with p and q corresponding densities, $\frac{dP}{dQ}$ the Radon-Nikodym derivative of P with respect to Q , and Ω the sample space. We also recall that a measure P is absolutely continuous with respect to Q ($P \ll Q$) if all nullsets of Q are nullsets of P . Lastly, if $X \sim p$ and $Y \sim q$ are random variables taking values in E , then:

$$E \left[\log \frac{p(X)}{q(Y)} \right] = D(p||q)$$

where p and q are the probability mass functions or densities of X and Y , depending on whether X and Y are continuous. It is important to keep in mind that while the Kullback-Leibler divergence is often thought of as a measure of difference between two distributions, it is not a metric, neither fulfilling symmetry, nor the triangle inequality:

$$\begin{aligned} D \left(\text{Ber} \left(\frac{1}{2} \right) \parallel \text{Ber} \left(\frac{3}{4} \right) \right) &\approx 0.21 \neq 0.19 \approx D \left(\text{Ber} \left(\frac{3}{4} \right) \parallel \text{Ber} \left(\frac{1}{2} \right) \right) \\ D \left(\text{Ber} \left(\frac{1}{4} \right) \parallel \text{Ber} \left(\frac{3}{4} \right) \right) &\approx 0.79 \not\leq 0.4 = 0.19 + 0.21 \approx D \left(\text{Ber} \left(\frac{1}{4} \right) \parallel \text{Ber} \left(\frac{1}{2} \right) \right) + D \left(\text{Ber} \left(\frac{1}{2} \right) \parallel \text{Ber} \left(\frac{3}{4} \right) \right) \end{aligned}$$

The KL divergence does, however, satisfy positive definiteness since if $X \sim p$ and $Y \sim q$ are random variables taking values in E :

$$\begin{aligned} D(p||q) &= E \left[\log \frac{p(X)}{q(X)} \right] = E \left[-\log \frac{q(x)}{p(X)} \right] > -\log \left(E \left[\frac{q(X)}{p(X)} \right] \right) = -\log \left(\int_E p(x) \frac{q(x)}{p(x)} dx \right) \\ &= -\log \left(\int_E q(x) dx \right) = -\log(1) = 0 \end{aligned}$$

due to Jensen's inequality since $-\log$ is strictly convex. If we have equality, we know that $\frac{p(X)}{q(X)}$ is constant. Because this constant gives zero under the map $-\log$, we have $\frac{p(X)}{q(X)} = 1$, thus, multiplying both sides of the equation by $q(X)$, we get: $p(X) = q(X)$.

5 Main Results Regarding Typicality

In the following theorem, we show that the probability of a sequence produced by a DMS can be expressed through the entropy of its type and the relative entropy of its type with respect to our generic distribution:

Theorem 1 [2]: for $x \in E^n$ produced according to a DMS with generic distribution Q and source alphabet E , $Q^n(x)$ depends only on x 's type P_x and is given by:

$$Q^n(x) = 2^{-n(H(P_x) + D(P_x||Q))} \quad (5.1)$$

Remark: in the following we use $\exp(\dots)$ to denote $2^{(\dots)}$

Proof:

$$\begin{aligned} Q^n(x) &= \prod_{i=1}^n Q(x_i) = \prod_{a \in E} Q(a)^{N(a|x)} = \prod_{a \in E} Q(a)^{nP_x(a)} = \exp \left(\sum_{a \in E} nP_x(a) \log Q(a) \right) = \\ &\exp \left(\sum_{a \in E} n(P_x(a) \log Q(a) + P_x(a) \log P_x(a) - P_x(a) \log P_x(a)) \right) = \\ &\exp \left(\sum_{a \in E} n \left(P_x(a) \log \frac{Q(a)}{P_x(a)} + P_x(a) \log P_x(a) \right) \right) = \\ &\exp \left(n \left(\sum_{a \in E} -P_x(a) \log \frac{P_x(a)}{Q(a)} + \sum_{a \in E} P_x(a) \log P_x(a) \right) \right) = \exp(n(-H(P_x) - D(P_x||Q))) \end{aligned}$$

□

It makes sense that the type of a sequence uniquely determines its probability, since the product measure $Q^n((X_1, \dots, X_n) = x) = Q^n((X_1, \dots, X_n) = \pi(x))$ is blind to any permutation of i.i.d. variables, and since the only information to be gained from having the exact sequence x as opposed to its type is x 's permutation. From this result we get a nice corollary if x has type Q :

Corollary 2 [2]: for $x \in E^n$ produced according to a DMS with generic distribution Q and source alphabet E , if x 's type is Q , then:

$$Q^n(x) = 2^{-nH(Q)} \quad (5.2)$$

Proof: Immediate by the positive definiteness of the Kullback-Leibler divergence after substituting Q for P_x in (5.1). $\square$

These results allow us to set bounds on the cardinalities of type classes:

Theorem 3 [2]: If $P \in \mathcal{P}_n$, then:

$$\frac{2^{nH(P)}}{(n+1)^{|E|}} \leq |T(P)| \leq 2^{nH(P)} \quad (5.3)$$

Remark: to find the exact value of $|T(P)|$ is actually not too difficult. It is simply the multinomial coefficient:

$$\binom{n}{N(a_1|x), \dots, N(a_{|E|}|x)}$$

where $\{a_i\}_{i=1}^{|E|}$ is an enumeration of E and x a sequence in $T(P)$. That this is indeed $T(P)$'s cardinality is obvious since all we wish to do, is arrange $N(a_1|x), \dots, N(a_{|E|}|x)$ in a sequence (not making distinctions between the $N(a_i|x)$ objects). By the definition of $N(a|x)$ we can also get an expression without our sequence x :

$$\binom{n}{nP(a_1), \dots, nP(a_{|E|})}$$

The problem with this value is that it is rather difficult to manipulate. We would much rather work with an exponent, leading to the motivation for the search of exponential bounds on $|T(P)|$.

We first observe that we can set a polynomial upper bound on $|\mathcal{P}_n|$ that, while very rough, will suit our needs:

Lemma 4 [2]:

$$|\mathcal{P}_n| \leq (n+1)^{|E|} \quad (5.4)$$

Proof (Lemma 4): One can easily observe that for a given $P \in \mathcal{P}_n$, P maps to $|E|$ elements and can assign each element a value in $\{\frac{i}{n}\}_{i=0}^n$. Hence:

$$|\mathcal{P}_n| \leq (n-0+1)^{|E|} = (n+1)^{|E|}$$

$\square$

Remark: Obviously, one can make this upper bound finer. For instance, if Ω is our sample space, one could notice that $1 = P(\Omega) = P\left(\bigcup_{i=1}^{|E|} \omega_i\right) = \sum_{i=1}^{|E|} P(\omega_i)$ since probability measures are σ -additive. Therefore, if we have assigned probabilities to all but one element, the last element must have probability $P(\omega_{|E|}) = 1 - \sum_{i=1}^{|E|-1} P(\omega_i)$, meaning we could set a better bound of $(n+1)^{|E|-1}$ on $|\mathcal{P}_n|$ and one could continue this process for all but two elements and so on. For us, however, it will only be important that this bound grows no faster than a polynomial as $n \rightarrow \infty$. Now for the proof of Theorem 3:

Proof (Theorem 3): for the upper bound on $|T(P)|$ we have:

$$1 \geq P^n(T(P)) = \sum_{x \in T(P)} P^n(x) \stackrel{(5.2)}{=} \sum_{x \in T(P)} 2^{-nH(P)} = |T(P)| 2^{-nH(P)}$$

Multiplying both sides by $2^{nH(P)}$, we get our upper bound. For the other direction, we will first prove that $P^n(T(Q))$ is maximized when $Q = P$. For this we consider the following quotient and show that it is greater than or equal to 1:

$$\begin{aligned} \frac{P^n(T(P))}{P^n(T(Q))} &= \frac{\sum_{x \in T(P)} P^n(x)}{\sum_{x \in T(Q)} P^n(x)} \stackrel{\text{i.i.d.}}{=} \frac{\sum_{x \in T(P)} \prod_{i=1}^n P(x_i)}{\sum_{x \in T(Q)} \prod_{i=1}^n P(x_i)} = \frac{\sum_{x \in T(P)} \prod_{a \in E} P(a)^{N(a|x)}}{\sum_{x \in T(Q)} \prod_{a \in E} P(a)^{N(a|x)}} = \\ \frac{\sum_{x \in T(P)} \prod_{a \in E} P(a)^{nP(a)}}{\sum_{x \in T(Q)} \prod_{a \in E} P(a)^{nQ(a)}} &= \frac{|T(P)| \prod_{a \in E} P(a)^{nP(a)}}{|T(Q)| \prod_{a \in E} P(a)^{nQ(a)}} \stackrel{\text{Rmk}}{=} \frac{\binom{n}{nP(a_1), \dots, nP(a_{|E|})}}{\binom{n}{nQ(a_1), \dots, nQ(a_{|E|})}} \prod_{a \in E} P(a)^{n(P(a)-Q(a))} = \\ &\prod_{a \in E} \frac{(nQ(a))!}{(nP(a))!} P(a)^{n(P(a)-Q(a))} \end{aligned}$$

We now use that $\frac{m!}{n!} \geq n^{m-n}$ since if $m \geq n$:

$$\frac{m!}{n!} = m(m-1) \cdots (m-n+1) \geq n^{m-n}$$

and if $m < n$:

$$\frac{m!}{n!} = \frac{1}{n(n-1) \cdots (n-m+1)} \geq \frac{1}{n^{n-m}} = n^{m-n}$$

Thus:

$$\begin{aligned} \prod_{a \in E} \frac{(nQ(a))!}{(nP(a))!} P(a)^{n(P(a)-Q(a))} &\geq \prod_{a \in E} (nP(a))^{n(Q(a)-P(a))} P(a)^{n(P(a)-Q(a))} = \prod_{a \in E} n^{n(Q(a)-P(a))} = \\ &n^{\sum_{a \in E} n(Q(a)-P(a))} = n^{n(1-1)} = 1 \end{aligned}$$

where in the second to last equality we observe that $\sum_{a \in E} P(a) = \sum_{a \in E} Q(a) = 1$ since P and Q are probability measures and we are summing over a partition of our source alphabet E . Knowing this, we can now argue (using the fact that our type classes partition E^n):

$$\begin{aligned} 1 &= \sum_{Q \in \mathcal{P}_n} P^n(T(Q)) \leq \sum_{Q \in \mathcal{P}_n} P^n(T(P)) \stackrel{(5.4)}{\leq} (n+1)^{|E|} P^n(T(P)) = (n+1)^{|E|} \sum_{x \in T(P)} P^n(x) \stackrel{(5.2)}{=} \\ &(n+1)^{|E|} \sum_{x \in T(P)} 2^{-nH(P)} = (n+1)^{|E|} |T(P)| 2^{-nH(P)} \end{aligned}$$

Multiplying both sides by $\frac{2^{nH(P)}}{(n+1)^{|E|}}$ yields our lower bound. $\square$

This leads us to a corollary that will later allow us to achieve the exponential bound referenced in the introduction:

Corollary 5 [2]: for distribution $P \in \mathcal{P}_n$ and distribution Q :

$$\frac{2^{-nD(P||Q)}}{(n+1)^{|E|}} \leq Q^n(T(P)) \leq 2^{-nD(P||Q)} \quad (5.5)$$

Proof:

$$Q^n(T(P)) = \sum_{x \in T(P)} Q^n(x) \stackrel{(5.1)}{=} \sum_{x \in T(P)} 2^{-n(H(P)+D(P||Q))} = |T(P)| 2^{-n(H(P)+D(P||Q))}$$

Plugging in our bounds for $|T(P)|$ from theorem 3, we get the desired inequalities. $\square$

6 Entropy as Uncertainty

We are finally in the position to discuss the motivation for entropy alluded to in section 2. It was mentioned that entropy is meant to be a measure of "uncertainty" of a random variable X but how can we describe, mathematically, what uncertainty means? As an example, if we take $E := \{0, 1\}$ as our source alphabet, we might consider $X \sim \text{Ber}(0.99)$ to be more certain than $Y \sim \text{Ber}(\frac{1}{2})$ since we can be more confident that X will be 1 than that Y will be any particular value. In fact, if we take uncertainty to mean unpredictability and try to predict the appearance of sequences $x, y \in \{0, 1\}^n$ produced according to $\text{Ber}(0.99)$ and $\text{Ber}(\frac{1}{2})$, respectively, we might notice that we can exclude a greater number of possible sequences in $\{0, 1\}^n$ from our prediction for x with more confidence: We can first see that, by the (strong) law of large numbers, for the random variable $Z \sim (\text{Ber}(p))^n$ and all $\epsilon > 0$:

$$0 = \Pr \left[\lim_{n \rightarrow \infty} \left(\frac{\sum_{i=1}^n Z_i}{n} \right) - E[\text{Ber}(p)] \geq \epsilon \right] = \Pr \left[\lim_{n \rightarrow \infty} \left(\frac{|\{i \in [1, n]_{\mathbb{N}} | Z_i = 1\}|}{n} \right) - p \geq \epsilon \right], \quad (6.1)$$

where we denote $\{1, 2, \dots, n\}$ with $[1, n]_{\mathbb{N}}$. So for large n , we get with probability close to 1 that about 99% of the elements in our sequence x and one half of those in y will be 1's, meaning the type P_x of x and P_y of y will be "about" equal to $\text{Ber}(0.99)$ and $\text{Ber}(\frac{1}{2})$, respectively. We will even show that for $n \rightarrow \infty$ and for every $\epsilon > 0$, $P_x \notin \{Q | D(Q || \text{Ber}(0.99)) > \epsilon\}$ with probability 1 (and similarly for y). This means, that in the limiting case, sequences in type classes other than $T(\text{Ber}(0.99))$ can be excluded as plausible predictions for x . This leaves us with the sequences in $T(\text{Ber}(0.99))$ as good choices for predictions about how x will look. Applying similar logic for y , we see that we are left with sequences in $T(\text{Ber}(\frac{1}{2}))$. Because, however, every sequence in $T(\text{Ber}(0.99))$ is equally likely to occur, we might consider the number of "good" choices $|T(\text{Ber}(0.99))|$ for predictions of x to be the "unpredictability" and therefore "uncertainty" about the appearance of x (similarly for y).

Throughout the last paragraph we have maintained a cursory assumption of n close to infinity to arrive at a correlation between the uncertainty of a sequence of our Bernoulli variables and the cardinalities of the type classes of their respective distributions. However, we need to somehow make this more rigorous since, if we require our measure of uncertainty to be additive, a perhaps desirable quality, we get an ill-defined object. If we try to average and obtain the average uncertainty of an element of the sequence, we get $\frac{|T(\cdot)|}{n}$, which diverges as n grows (we average because considering sequences $x \in E^n$ corresponds to the uncertainty of the random variable $X \sim P^n$ whereas considering elements corresponds to that of the random variable $X \sim P$). This becomes obvious after observing that the sizes of the type classes $|T(\text{Ber}(0.99))|$ and $|T(\text{Ber}(\frac{1}{2}))|$, for instance, explode according to their respective multinomial (in this case binomial) coefficients as $n \rightarrow \infty$ and these values are of course not in $O(n)$. Hence, we would like to find a function that grows with $|T(P)|$ where we keep n fixed while still tempering $|T(P)|$'s growth as $n \rightarrow \infty$, such that our new value is in $O(n)$ and we can average. Because we know $|T(P)|$ grows exponentially, we might consider a logarithm, leading to $\frac{\log |T(P)|}{n}$ as our uncertainty measure. But now, because $\frac{\log(\cdot)}{n}$ is strictly monotonically increasing, by Theorem 3:

$$\frac{\log \left(\frac{2^{nH(P)}}{(n+1)^{|E|}} \right)}{n} \leq \frac{\log |T(P)|}{n} \leq \frac{\log(2^{nH(P)})}{n}$$

since strictly monotonically increasing functions do not alter inequalities. The right side simplifies to the entropy of P and:

$$\frac{\log \left(\frac{2^{nH(P)}}{(n+1)^{|E|}} \right)}{n} = H(P) - |E| \frac{\log(n+1)}{n} \quad (6.2)$$

We now see that, for $n \rightarrow \infty$, the term on the right of the right hand side of (6.2) converges to 0, meaning our inequalities sandwich to the entropy $H(P)$ when we take limits and:

$$\lim_{n \rightarrow \infty} \frac{\log |T(P)|}{n} = H(P) \quad (6.3)$$

In this case, some authors write $|T(P)| \doteq 2^{nH(P)}$, to say $|T(P)|$ and $2^{nH(P)}$ "agree to first order in the exponent". More specifically [2]:

$$a_n \doteq b_n \iff \lim_{n \rightarrow \infty} \frac{1}{n} \log \frac{a_n}{b_n} = 0 \quad (6.4)$$

This means that the measure of uncertainty we proposed after some logical considerations about how it should act is exactly the entropy we defined in section 2. This interpretation of entropy is tightly connected to the asymptotic equipartition property, which will be discussed in the next section. In the meantime, let us prove the claim made in the paragraph below (6.1) that for $n \rightarrow \infty$ and for every $\epsilon > 0$, $P_x \notin \{Q|D(Q||\text{Ber}(0.99)) > \epsilon\}$ with probability 1. For this we will need the notion of an (ϵ) -typical set:

$$T_Q^{\epsilon,n} := \{x \in E^n | D(P_x||Q) \leq \epsilon\}$$

which represents the sequences x with type diverging no more than ϵ from Q with respect to the Kullback-Leibler divergence. This means that the sequences we mentioned earlier can be represented by the set $(T_{\text{Ber}(0.99)}^{\epsilon})^c$. As in the previous sentence, the n in the superscript is sometimes omitted. The sequences in the typical set are called ϵ -weakly typical (or just "weakly typical"), however, since this thesis will not discuss strong typicality [3], which also requires finite support, we will simply call them typical/ ϵ -typical. Now to prove the statement we made:

Theorem 6 [2]: For all $\epsilon > 0$ and distributions Q :

$$Pr((T_Q^{\epsilon})^c) \leq (n+1)^{|E|} 2^{-n\epsilon}$$

Proof:

$$\begin{aligned} Pr((T_Q^{\epsilon,n})^c) &= \sum_{P \in \mathcal{P}_n: D(P||Q) > \epsilon} Q^n(T(P)) \stackrel{(5.5)}{\leq} \sum_{P \in \mathcal{P}_n: D(P||Q) > \epsilon} 2^{-nD(P||Q)} \leq \sum_{P \in \mathcal{P}_n: D(P||Q) > \epsilon} 2^{-n\epsilon} \stackrel{(5.4)}{\leq} \\ &\quad (n+1)^{|E|} 2^{-n\epsilon} \end{aligned}$$

□

That this implies our claim that $\lim_{n \rightarrow \infty} Pr((T_Q^{\epsilon,n})^c) = 0$ is clear since 2^{-n} converges faster than any polynomial diverges. We get the following corollary:

Corollary 7 [2]: let $X_1, X_2, \dots$ be a DMS with generic distribution Q , then:

$$D(P_{X^n}||Q) \rightarrow 0 \text{ almost surely (a.s.)}$$

where P_{X^n} is the type of the random sequence $(X_1, X_2, \dots, X_n)$.

Proof: we have:

$$\sum_{n \in \mathbb{N}} Pr(D(P_{X^n}||Q) > \epsilon) = \sum_{n \in \mathbb{N}} Pr((T_Q^{\epsilon,n})^c) \leq \sum_{n \in \mathbb{N}} 2^{-n(\epsilon - |E| \frac{\log(n+1)}{n})} \quad (6.5)$$

and:

$$\limsup_{n \rightarrow \infty} \left(2^{-n(\epsilon - |E| \frac{\log(n+1)}{n})} \right)^{\frac{1}{n}} = \limsup_{n \rightarrow \infty} 2^{-\frac{n\epsilon}{n}} (n+1)^{\frac{|E|}{n}} = 2^{-\epsilon}$$

is strictly less than 1 for all fixed $\epsilon > 0$. In the last equality we used that for all polynomials $p \in \mathbb{R}[x]$, $\lim_{n \rightarrow \infty} |p(n)|^{\frac{1}{n}} = 1$ where this can be verified by applying L'Hôpital's rule [8] to the logarithm of the limit and then re-exponentiating. We can now use Cauchy's root test to conclude that our series converges. We recall the first Borel-Cantelli lemma [6, 7]:

$$\sum_{n \in \mathbb{N}} Pr(E_n) < \infty \implies Pr\left(\limsup_{n \rightarrow \infty} E_n\right) = 0$$

where $E_1, E_2, \dots$ are events. In words, if the expected number of occurrences of the events E_i is finite, then the actual number of occurrences is also finite almost surely. Since our sum in (6.5) is finite, the Borel-Cantelli lemma applies and: $Pr(\limsup_{n \rightarrow \infty} [D(P_X || Q) > \epsilon]) = 0$. Therefore, $D(P_{X^n} || Q) \xrightarrow{a.s.} 0$. $\square$

7 The Asymptotic Equipartition Property

In corollary 7 we saw that the type P_X of a random sequence produced by a DMS becomes increasingly deterministic in the sense that P_X looks more and more like our generic distribution P with probability 1. We also saw in theorem 1 that $Q^n(x) = 2^{-n(H(P_x) + D(P_x || Q))}$ but because the second term in the exponent goes to zero, one would expect the probability to approach an expression depending only on $H(Q)$. This is, in fact, a cardinal result in information theory:

Theorem 8, Asymptotic Equipartition Property (AEP) [9]: if $X_1, X_2, \dots$ is a DMS with generic distribution Q , then:

$$-\frac{1}{n} \log Q^n(X_1, X_2, \dots, X_n) \xrightarrow{a.s.} H(Q) \quad (7.1)$$

Remark: this follows immediately from Corollary 7, (5.1), (5.2) and (6.3) but because this result has such an intimate connection to the law of large numbers (in fact, it is often called "the law of large numbers for information theory"), we illustrate this connection in our proof:

Proof: as functions of i.i.d. variables:

$$Y_i := -\log Q(X_i)$$

Y_i are also i.i.d. and:

$$\begin{aligned} -\frac{1}{n} \log Q^n(X_1, X_2, \dots, X_n) &= -\frac{1}{n} \log(Q(X_1)Q(X_2) \cdots Q(X_n)) = \\ &= -\frac{1}{n} (\log Q(X_1) + \log Q(X_2) + \cdots + \log Q(X_n)) = \bar{Y}_n, \end{aligned}$$

where $\bar{Y}_n$ denotes the empirical mean of the first n Y_i . Now, by the strong law of large numbers [10]:

$$\bar{Y}_n \xrightarrow{a.s.} E[Y_1] = -E[\log(X_1)] = H(X) = H(Q)$$

$\square$

Remark: one can also show convergence in the first mean (Shannon-McMillan-Breiman theorem [11]):

$$E \left[\left| \lim_{n \rightarrow \infty} -\frac{1}{n} \log Q^n(X_1, X_2, \dots, X_n) - H(Q) \right| \right] = 0$$

but this thesis abstains from the proof of this result as it is outside of our scope.

8 Sanov's Inequality and Large Deviation Theorem

The exponential bound mentioned in the introduction follows easily from Corollary 5 and Theorem 6:

Theorem 9, Sanov's inequality [2, 17]: let $X_1, X_2, \dots$ be a DMS with generic distribution Q and source alphabet E . Further, let $\mathcal{A}$ be a set of distributions, then:

$$Q^n(\cup_{P \in \mathcal{A} \cap \mathcal{P}_n} T(P)) \leq (n+1)^{|E|} 2^{-nD(P^* || Q)} \quad (8.1)$$

where $P^* := \operatorname{argmin}_{P \in \mathcal{A}} D(P || Q)$

Remarks:

- If $Q \in \mathcal{A}$, the inequality becomes uninteresting as it simplifies to a probability being bounded by a polynomial. We therefore assume from now on, that Q is never in $\mathcal{A}$.
- We write $Q^n(T(\mathcal{A} \cap \mathcal{P}_n))$ to mean $Q^n(\cup_{P \in \mathcal{A} \cap \mathcal{P}_n} T(P))$. We may further simplify this to $Q(T(\mathcal{A}))$.
- With $cl(\mathcal{A})$ we mean the closure of $\mathcal{A}$ with respect to the weak topology on probability measures. See the discussion in the paragraph before Theorem 10.

Proof: see the proof of Theorem 6 and set $\epsilon := D(P^*||Q)$.

Interpretation: this inequality tells us that the probability that $x \in E^n$ is in a given set B can be given an upper bound depending only on the element y with type "closest" to the generic distribution Q with respect to the Kullback-Leibler divergence. Furthermore if y 's type is not the generic distribution Q then we have a probability that decays exponentially in n . In Sanov's large deviation theorem we will even see that this probability and its bound agree to first order in the exponent (6.4) if a certain condition on $\mathcal{A}$ is met. This agreement means that we cannot find any better exponent than $-nD(P^*||Q)$ in (8.1).

The aforementioned condition will be that " $\mathcal{A}$ is the closure of its interior" with respect to the "weak topology" on our probability measures. The weak topology is essentially the weakest (smallest) topology on the space of our probability measures $\mathcal{P}(\Omega)$, such that all functionals in our dual space $\mathcal{P}(\Omega)^*$ remain continuous. This also has a connection to the Lévy-Prokhorov metric but this thesis will not go into any more detail. Please see [12, 13]. Essentially what this condition allows us to do, is ensure that P^* lies in $\mathcal{A}$ because $\mathcal{A}$ is closed, and that $\mathcal{A}$ is nonempty, such that every distribution in $\mathcal{A}$ is can be approximated through others also in $\mathcal{A}$ (see proof below).

Theorem 10, Sanov's large deviation theorem [2, 17]: let $X_1, X_2, \dots$ be a DMS with generic distribution Q and source alphabet E . Further, let $\mathcal{A}$ be a set of distributions with $\mathcal{A} = cl(\mathring{\mathcal{A}})$ with respect to the weak topology, then:

$$\lim_{n \rightarrow \infty} \log \frac{Q^n(T(\mathcal{A}))}{n} = -D(P^*||Q)$$

Proof:

$$Q^n(T(\mathcal{A})) = \sum_{P \in \mathcal{A} \cap \mathcal{P}_n} Q^n(T(P)) \geq Q^n(T(P_n)) \stackrel{(5.5)}{\geq} \frac{2^{-nD(P_n||Q)}}{(n+1)^{|E|}} \quad (8.2)$$

where, since P^* can be approximated through distributions in $\mathcal{A} \setminus \{P^*\}$ (because $\mathcal{A} = cl(\mathring{\mathcal{A}})$), and because $\cup_{n \in \mathbb{N}} \mathcal{P}_n$ is dense in the set of all probability distributions, we can find $P_i \in \mathcal{P}_i$ for every i such that $\lim_{i \rightarrow \infty} P_i = P^*$. Thus, since:

$$\lim_{n \rightarrow \infty} \frac{\log \frac{2^{-nD(P_n||Q)}}{(n+1)^{|E|}}}{n} = -D(P^*||Q) = \lim_{n \rightarrow \infty} \frac{\log 2^{-nD(P_n||Q)}}{n}$$

we see that our upper bound (8.1) and our lower bound (8.2) agree to first order in the exponent. $\square$

Going back to our example with $Ber(\frac{1}{2})$, it's logical that the probability of a set $B := \{x \in \cup_{n \in \mathbb{N}} E^n \mid \text{at least three quarters of } x \text{ are } 1\text{'s}\}$, for instance, becomes increasingly less dependent on sequences x with a ratio of $\frac{3}{4} + \epsilon$ of 1's to 0's, since their probabilities decay exponentially quicker than those with a ratio of $\frac{3}{4}$ (the types of the former have a larger Kullback-Leibler divergence with respect to $Ber(\frac{1}{2})$, see: (5.1)).

9 Examples of Sanov's inequality

Sanov's inequality allows us to revert our problem of searching for a bound on a probability to a concrete minimization problem, which, in many cases, is more manageable of a problem. In many applications one wishes to bound a probability of the form [2]:

$$Pr \left(\bigcap_{j=1}^k \left(\frac{1}{n} \sum_{i=1}^n f_j(X_i) \geq c_j \right) \right) \quad (9.1)$$

where f_j are given real-valued functions ($\mathbb{R} \rightarrow \mathbb{R}$, c_j given constants, and $X_1, X_2, \dots, X_n$ random variables produced according to the DMS with generic distribution Q and source alphabet E . For a given sequence x with type $P_x \in \mathcal{P}_n$, we notice that the sum in (9.1) is no different than:

$$E_{P_x}[f_j(Y)] = \sum_{a \in E} P_x(a) f_j(a) = \sum_{a \in E} \frac{N(a|x)}{n} f_j(a) = \frac{1}{n} \sum_{i=1}^n f_j(x_i) = (9.1)$$

where $E_{P_x}[\cdot]$ is the expected value with respect to the measure corresponding to P_x and $Y \sim P$ a random variable with corresponding mass function P_x . Therefore, a sequence $x \in E^n$ is inside the set¹ in (9.1) if and only if its type P_x satisfies $\sum_{a \in E} P_x(a) f_j(a) \geq c_j$. Therefore, in the framework of Theorem 9 (Sanov's inequality), our set $\mathcal{A}$ can be written as:

$$\mathcal{A} := \left\{ P \in \mathcal{P} \left| \forall j \in [1, k]_{\mathbb{N}} \sum_{a \in E} P(a) f_j(a) \geq c_j \right. \right\} \quad (9.2)$$

where $\mathcal{P} := \cup_{i \in \mathbb{N}} \mathcal{P}_n$. To find Sanov's bound on (9.1), Sanov's inequality tells us that we first need to find P^* , the closest type in $\tilde{\mathcal{A}} = \mathcal{A}$ to the underlying generic distribution Q . This means we have to find the type $P^* \in \mathcal{A}$ that minimizes the Kullback-Leibler divergence. We first realize that, because $Q \notin \mathcal{A}$, P^* will satisfy our constraints with equality. Therefore, we can use Lagrange multipliers [14] to minimize the Kullback-Leibler divergence on the boundary of $\mathcal{A}$, where equality is attained. Our minimum-achieving mass function will then also be the minimizer over all of the mass functions in $\mathcal{A}$, since those in the interior $\mathring{\mathcal{A}} = \mathcal{A} \setminus \partial \mathcal{A}$ will have greater Kullback-Leibler divergence than those on the boundary. Thus, we construct the following functional:

$$\begin{aligned} \mathcal{F}(P) &:= D(P||Q) + \sum_{j=1}^k \lambda_j \sum_{a \in E} P(a) f_j(a) + \nu \sum_{a \in E} P(a) = \\ &\sum_{a \in E} P(a) \log \frac{P(a)}{Q(a)} + \sum_{j=1}^k \lambda_j \sum_{a \in E} P(a) f_j(a) + \nu \sum_{a \in E} P(a) \end{aligned}$$

in which our constraints in (9.2) are reflected. Additionally, the last term represents the constraint by the fact that P must be a probability measure. In other words: $\sum_{a \in E} P(a) \stackrel{!}{=} 1$.

We would now like to find partial derivatives with respect to $P(y)$ in order to find for what values $P(a)$ we have that $\mathcal{F}$ has an extremum. This partial differentiation with respect to $P(y)$ means we are regarding $\mathcal{F}$ as a function solely dependent on $P(y)$. This means, for all other $a \in E \setminus \{y\}$, $P(a)$ is treated as a constant, since the output of a function evaluated at one point does not change with the output evaluated at another, meaning $\frac{\partial P(a)}{\partial P(y)} = 0$. This is because, at this point, P is still a general function. Of course, for probability distributions, there is dependence between the $P(a)$, as they must sum to one, but we can always correct a posteriori whatever value the sum yields by adjusting ν appropriately. Therefore, by linearity of the partial derivative:

$$\begin{aligned} \frac{\partial \mathcal{F}}{\partial P(y)} &= \frac{\partial}{\partial P(y)} \left[\sum_{a \in E} P(a) \log \frac{P(a)}{Q(a)} + \sum_{j=1}^k \lambda_j \sum_{a \in E} P(a) f_j(a) + \nu \sum_{a \in E} P(a) \right] = \\ &\sum_{a \in E} \frac{\partial}{\partial P(y)} \left[P(a) \log \frac{P(a)}{Q(a)} \right] + \sum_{j=1}^k \lambda_j \sum_{a \in E} \frac{\partial}{\partial P(y)} [P(a) f_j(a)] + \nu \sum_{a \in E} \frac{\partial}{\partial P(y)} [P(a)] = \\ &\sum_{a \in E} \left(\delta_{a,y} \left(Q(a) + \log \frac{P(a)}{Q(a)} \right) \right) + \sum_{j=1}^k \lambda_j \sum_{a \in E} \delta_{a,y} f_j(a) + \nu \sum_{a \in E} \delta_{a,y} = \end{aligned}$$

¹which is inside the probability function

$$1 + \log \frac{P(y)}{Q(y)} + \sum_{j=1}^k \lambda_j f_j(y) + \nu \quad (9.3)$$

where $\delta_{a,y}$ is the Kronecker delta:

$$\delta_{a,y} := \begin{cases} 1, & a = y \\ 0, & \text{otherwise} \end{cases}$$

Setting (9.3) to zero and solving for $P(y)$, we attain:

$$P(y) = Q(y) 2^{-\nu-1-\sum_{j=1}^k \lambda_j f_j(y)}$$

We can also notice that dividing our expression by the sum of all $P(y)$'s ensures our constraint that P be a probability mass function, thereby eliminating the need for ν :

$$\begin{aligned} P(y) &= \frac{Q(y) 2^{-\nu-1} 2^{-\sum_{j=1}^k \lambda_j f_j(y)}}{\sum_{a \in E} Q(a) 2^{-\nu-1} 2^{-\sum_{j=1}^k \lambda_j f_j(a)}} = \frac{Q(y) 2^{-\nu-1} 2^{-\sum_{j=1}^k \lambda_j f_j(y)}}{2^{-\nu-1} \sum_{a \in E} Q(a) 2^{-\sum_{j=1}^k \lambda_j f_j(a)}} \implies \\ P(y) &= \frac{Q(y) 2^{-\sum_{j=1}^k \lambda_j f_j(y)}}{\sum_{a \in E} Q(a) 2^{-\sum_{j=1}^k \lambda_j f_j(a)}} \end{aligned} \quad (9.4)$$

Applying this method to every $P(y)$ yields us $|E|$ equations with the above form. Combining this with our k constraint equations, we have $|E| + k$ equations and $|E| + k$ unknowns, meaning we can solve the system of equations for each $P(a)$ and choose the λ_j 's such that they satisfy our constraints (this is usually done numerically). Lastly, taking the second variation of (9.3), plugging in (9.4) and showing its positivity proves that this extremum is indeed a minimum. For the sake of brevity, however, we omit these calculations with an alternative proof:

$$\begin{aligned} D(P||Q) &= \sum_{a \in E} P(a) \log \frac{P(a)}{Q(a)} = \sum_{a \in E} P(a) \log \frac{P(a)P^*(a)}{P^*(a)Q(a)} = \sum_{a \in E} P(a) \left(\log \frac{P(a)}{P^*(a)} + \log \frac{P^*(a)}{Q(a)} \right) = \\ &= \sum_{a \in E} P(a) \log \frac{P(a)}{P^*(a)} + \sum_{a \in E} P(a) \log \frac{P^*(a)}{Q(a)} = D(P||P^*) + \sum_{a \in E} P(a) \log \frac{P^*(a)}{Q(a)} \stackrel{(i)}{\geq} \sum_{a \in E} P(a) \log \frac{P^*(a)}{Q(a)} = \\ &= \sum_{a \in E} P(a) \log P^*(a) - \sum_{a \in E} P(a) \log Q(a) \stackrel{(9.4)}{=} \sum_{a \in E} P(a) \log \left(\frac{Q(a) 2^{-\sum_{j=1}^k \lambda_j f_j(a)}}{\sum_{b \in E} Q(b) 2^{-\sum_{j=1}^k \lambda_j f_j(b)}} \right) - \sum_{a \in E} P(a) \log Q(a) = \\ &= \sum_{a \in E} P(a) \log 2^{-\sum_{j=1}^k \lambda_j f_j(a)} + \sum_{a \in E} P(a) \log Q(a) - \sum_{a \in E} P(a) \log \left(\sum_{b \in E} Q(b) 2^{-\sum_{j=1}^k \lambda_j f_j(b)} \right) - \sum_{a \in E} P(a) \log Q(a) = \\ &= \sum_{a \in E} P(a) \left(-\sum_{j=1}^k \lambda_j f_j(a) \right) - \log \left(\sum_{b \in E} Q(b) 2^{-\sum_{j=1}^k \lambda_j f_j(b)} \right) \sum_{a \in E} P(a) \stackrel{(ii)}{=} \\ &= \sum_{a \in E} P^*(a) \left(-\sum_{j=1}^k \lambda_j f_j(a) \right) - \log \left(\sum_{b \in E} Q(b) 2^{-\sum_{j=1}^k \lambda_j f_j(b)} \right) \sum_{a \in E} P^*(a) = \\ &= \sum_{a \in E} P^*(a) \log 2^{-\sum_{j=1}^k \lambda_j f_j(a)} - \sum_{a \in E} P^*(a) \log \left(\sum_{b \in E} Q(b) 2^{-\sum_{j=1}^k \lambda_j f_j(b)} \right) + \sum_{a \in E} P^*(a) \log Q(a) - \sum_{a \in E} P^*(a) \log Q(a) = \\ &= \sum_{a \in E} P^*(a) \log \left(\frac{Q(a) 2^{-\sum_{j=1}^k \lambda_j f_j(a)}}{\sum_{b \in E} Q(b) 2^{-\sum_{j=1}^k \lambda_j f_j(b)}} \right) - \sum_{a \in E} P^*(a) \log Q(a) = \\ &= \sum_{a \in E} P^*(a) \log P^*(a) - \sum_{a \in E} P^*(a) \log Q(a) = \sum_{a \in E} P^*(a) \log \frac{P^*(a)}{Q(a)} = D(P^*||Q) \end{aligned}$$

where P is any other distribution satisfying the constraints.

Remarks

- (i) follows from the non-negativity of the Kullback-Leibler divergence.
- (ii) stems from the fact that both P and P^* are probability measures that also satisfy our constraints. Hence, $\sum_{a \in E} P(a) \left(-\sum_{j=1}^k \lambda_j f_j(a) \right) = -\sum_{j=1}^k \lambda_j \sum_{a \in E} P(a) f_j(a) = -\sum_{j=1}^k c_j = -\sum_{j=1}^k \lambda_j \sum_{a \in E} P^*(a) f_j(a) = \sum_{a \in E} P^*(a) \left(-\sum_{j=1}^k \lambda_j f_j(a) \right)$ and $\sum_{a \in E} P(a) = 1 = \sum_{a \in E} P^*(a)$.
- Lastly, we see that we have equality if and only if the Kullback-Leibler divergence of P with respect to P^* is zero, which holds if and only if $P = P^*$ due to the positive-definiteness of the Kullback-Leibler divergence. Therefore, as an additional result, we see that our minimizer P^* is unique.
- One can show an analogous result for continuous random variables. One need only replace sums with integrals and mass functions with densities. However, we no longer have a unique minimizing density, rather only a unique equivalence class of densities where our equivalence relation is almost sure equality. In other words, the minimizing density is uniquely defined up to a null set (that is, a set of Lebesgue measure zero [15]). This is because, in the continuous case, the Kullback-Leibler divergence is positive semi-definite, blind to differences of density functions on null sets.

As a simple example, we can consider the probability that n rolls of a fair die yields an empirical mean greater than or equal to four [2]. In this case our source alphabet is $E := \{1, 2, 3, 4, 5, 6\}$, our generic distribution Q is uniform and $\mathcal{A}$ can be written as:

$$\mathcal{A} := \left\{ P \in \mathcal{P} \left| \sum_{i=1}^6 iP(i) \geq 4 \right. \right\}$$

Therefore we have one constraint. According to (9.4):

$$P^*(y) = \frac{\frac{1}{6}2^{-\lambda y}}{\sum_{a \in E} \frac{1}{6}2^{-\lambda a}} = \frac{2^{-\lambda y}}{\sum_{a \in E} 2^{-\lambda a}}$$

Solving numerically, we get $\lambda \approx 0.25$, $P^*(1) \approx 0.10$, $P^*(2) \approx 0.12$, $P^*(3) \approx 0.15$, $P^*(4) \approx 0.17$, $P^*(5) \approx 0.21$, and $P^*(6) \approx 0.25$. Plugging P^* into $D(P^*||Q)$, we get $D(P^*||Q) \approx 0.06$. Thus, Sanov's inequality allows us to conclude: $P(T(\mathcal{A})) \leq (n+1)^6 2^{-0.06n}$.

Another example is a fair coin [2]. We have source alphabet $E := \{0, 1\}$ and uniform distribution $Q \sim \text{Ber}(\frac{1}{2})$. If we want to find the probability that our sequence consists of more than 70% heads, for instance, then:

$$\mathcal{A} := \left\{ P \in \mathcal{P} \left| \frac{P(1) + P(0)}{2} \geq 0.7 \right. \right\}$$

Obviously, the maximizing P^* will be $P^*(1) = 0.7$ and $P^*(0) = 0.3$, which yields $D\left(P^* \left\| \text{Ber}\left(\frac{1}{2}\right)\right.\right) = 0.119$, meaning: $P(T(\mathcal{A})) \leq (n+1)^2 2^{-0.119n}$.

Remark: in the above examples, one sees the limitations of our inequality if n is not big. For instance, our first inequality only falls below 1 at $n = 997$ and our second at $n = 116$. While finer bounds, asymptotically (end of the paragraph after the proof of Sanov's inequality), make little difference, they can serve as better bounds for smaller n . An example of these adjustments can be seen in Chernoff bounds, which are not in the scope of this thesis [16]. One can also make the inequality tighter by adjusting the rather rough inequality in Lemma 4 to be finer (this new tighter inequality can replace the old one used in the proof of Sanov's inequality). Nevertheless, in a very vast majority of applications, one is working with large n . Furthermore, in situations where performing an experiment/trial is resource-expensive, a finer estimate is vital to an efficient solution. Whether it be gauging the probability of the occurrence of adverse effects of a new drug, or the probability that the material making up a bridge defects, a simple application of Sanov's inequality yields a tight upper bound.

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